

Suggested Solutions to Exercises 10

1. For a non-dividend paying stock, put-call parity gives at a time t

$$p + S = c + Xe^{-r(T-t)}$$

where T is the expiry date, and c and p are the prices of a call and a put option, respectively.

- (a) Differentiating with respect to S :

$$\frac{\partial p}{\partial S} + 1 = \frac{\partial c}{\partial S}$$

This shows that the Delta of a European put equals the Delta of the corresponding European call minus 1.

- (b) Differentiating with respect to S again:

$$\frac{\partial^2 p}{\partial S^2} = \frac{\partial^2 c}{\partial S^2}$$

Hence the Gamma of a European put equals the Gamma of a European call.

- (c) Differentiating the put-call parity relationship with respect to σ :

$$\frac{\partial p}{\partial \sigma} = \frac{\partial c}{\partial \sigma}$$

showing that the Vega of a European put equals the Vega of a European call.

- (d) Finally, differentiating the put-call parity equation with respect to t we obtain the following relationship between the Thetas:

$$\frac{\partial p}{\partial t} = rXe^{-r(T-t)} + \frac{\partial c}{\partial t}$$

2. We need to differentiate the expression for a call option, C say, with respect to S . The Black-Scholes formula gives

$$C = SN(d_1) - Xe^{-r(T-t)}N(d_2)$$

so that

$$\begin{aligned}\Delta &= \frac{\partial C}{\partial S} \\ &= N(d_1) + SN'(d_1)\frac{\partial d_1}{\partial S} - Xe^{-r(T-t)}N'(d_2)\frac{\partial d_2}{\partial S} \\ &= N(d_1) + (SN'(d_1) - Xe^{-r(T-t)}N'(d_2))\frac{\partial d_1}{\partial S}\end{aligned}$$

where we have used the fact that

$$d_2 = d_1 - \sigma\sqrt{T-t} \quad \text{so that} \quad \frac{\partial d_1}{\partial S} = \frac{\partial d_2}{\partial S}.$$

Now use the density of a standard normal distribution

$$N'(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$$

to give

$$\begin{aligned}N'(d_2) &= \frac{1}{\sqrt{2\pi}}e^{-d_2^2/2} = \frac{1}{\sqrt{2\pi}}e^{-d_1^2/2} \exp\left\{\sigma\sqrt{T-t}d_1 - \sigma^2(T-t)/2\right\} \\ &= N'(d_1) \exp\left\{\log\left(\frac{S}{X}\right) + \left(r + \frac{\sigma^2}{2}\right)(T-t) - \frac{\sigma^2}{2}(T-t)\right\} \\ &= N'(d_1) \frac{S}{X} e^{r(T-t)}\end{aligned}$$

where in the first step above we have expanded $d_2^2 = (d_1 - \sigma\sqrt{T-t})^2$, and then used the result

$$d_1 = \frac{\log(\frac{S}{X}) + \left(r + \frac{\sigma^2}{2}\right)(T-t)}{\sigma\sqrt{T-t}}$$

from the Black-Scholes formula to obtain the equality in the second step. Re-arranging the final expression in the third step, we have

$$SN'(d_1) = Xe^{-r(T-t)}N'(d_2),$$

hence returning to the expression for Δ in the previous page, we have $\Delta = N(d_1)$.

3. From the results of questions 1(a) and 2, we have that the Delta for a put is $\Delta_C - 1 = N(d_1) - 1$, where Δ_C is the Delta of a corresponding call option.
4. Suppose we add x_F of F and x_G of G to the portfolio. For the new portfolio to be delta neutral we need

$$\Delta_P + x_F \Delta_F + x_G \Delta_G = 0$$

which gives

$$2 - x_F + 2x_G = 0.$$

For the portfolio to be gamma neutral as well we need

$$\Gamma_P + x_F \Gamma_F + x_G \Gamma_G = 0$$

which gives

$$3 + 2x_F - 2x_G = 0.$$

These two simultaneous equations solve to give

$$x_F = 1 \quad \text{and} \quad x_G = -1/2$$

so that we go long one F and short half G .

5. We have $\Delta_S = 1$ and $\Gamma_S = 0$ so that the equations from question 4 become

$$2 - x_F + x_S = 0$$

and

$$3 + 2x_F = 0.$$

These solve to give

$$x_F = -3/2 \quad \text{and} \quad x_S = 7/2.$$

6. A hedging portfolio will be a long position in a replicating portfolio which will consist of shares and the riskless asset. Call this portfolio value Π_t , then we have $\Pi_t = \Delta S_t + \psi_t B_t$ for some amount ψ_t , where we have used the standard result for hedging derivatives that an amount Δ of the stock must be used. B_t is the value of the riskless asset, and ψ_t is the amount held, so that we are interested in determining the

product of these two for the total value, $R_t = \psi_t B_t$ say. Therefore we have that $R_t = \Pi_t - \Delta S_t$. The value process of the derivative, f_t say, must follow the BSM PDE, which can be expressed in terms of the Greeks as follows:

$$\begin{aligned} r f_t &= r \frac{\partial f}{\partial S} S_t + \frac{\partial f}{\partial t} + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} \sigma^2 S^2 \\ &= r \Delta S_t + \Theta + \frac{1}{2} \Gamma \sigma^2 S^2 \end{aligned}$$

As $\Pi_t = f_t$, we have the result

$$r(R_t + \Delta S_t) = r \Delta S_t + \Theta + \frac{1}{2} \Gamma \sigma^2 S_t^2$$

which gives

$$R_t = \frac{1}{r} \left[\Theta + \frac{1}{2} \Gamma \sigma^2 S_t^2 \right]$$

This is a long position in this amount R_t .